Project Proposal (Proposed Solution) Report

Date	1 July 2026
Team ID	—
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

This proposal report describes how Rising Waters uses machine learning to give residents and disaster-management teams a fast, data-driven flood risk assessment, replacing slow and generic manual warnings.

Project Overview

Field	Details
Objective	Provide an instant, data-driven Flood / No-Flood risk prediction from live weather and terrain readings, so at-risk communities can act early.
Scope	End-to-end pipeline covering dataset generation, EDA, preprocessing, multi-model training/selection, and a Flask web app for real-time predictions.

Problem Statement

Field	Details
Description	Existing flood warnings are generic and slow, and residents lack a simple tool that combines rainfall, river level, drainage quality and terrain into one clear risk signal.
Impact	Delayed or unclear warnings increase risk to life and property; a fast, understandable prediction tool improves preparedness and response time.

Proposed Solution

Field	Details
Approach	Train and compare multiple classifiers (Decision Tree, Random Forest, KNN, XGBoost) on historical flood-related data, automatically select the most accurate, and serve it through a Flask web application.
Key Features	• Multi-model training with automatic best-model selection (XGBoost, ~75.3% test accuracy) • Web form covering 10 weather/terrain features with sensible defaults and ranges • Instant Flood / No-Flood prediction with probability score • Dedicated result pages with safety guidance for flood-risk cases

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU used for model training (no GPU needed)	Standard multi-core CPU
Memory	RAM specification	8 GB
Storage	Disk space for data, models and logs	< 1 GB
Framework	Python web framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, joblib, xgboost
Development Environment	IDE	VS Code / Jupyter / PyCharm
Data Source, size, format	Flood-related weather & terrain dataset	data/flood_data.csv (simulated, structured like a real Kaggle flood dataset)